

How Fintech and Digital Currencies are Reshaping Deposit-Taking Institutions and Payment Systems

FINN1051 – Global Financial Systems and Markets

I. Background and Theoretical Context

The global financial system is undergoing a profound transformation. Fintech and digital currencies are reshaping the architecture of deposit-taking institutions and payment systems in ways that challenge decades of established financial theory and practice. At its core, the theory of financial intermediation holds that institutions such as banks exist to reduce transaction costs and to mitigate the asymmetric information that arises between savers and borrowers (Mishkin and Eakins, 2024). For much of the twentieth century, this function was performed almost exclusively by regulated commercial banks and building societies. That dominance is now under serious challenge.

Fintech firms leverage artificial intelligence, open banking data, and mobile infrastructure to perform the same intermediation functions at a fraction of the traditional cost. The UK alone is home to over 3,500 fintech companies spanning digital banking, payments, wealth management, and embedded finance, generating revenues of approximately £32.4 billion in 2024, with projected annual growth of 16.1% to 2032 (Innovate Finance, 2024). Cross-border payments, historically slow, expensive, and reliant on chains of correspondent banks, can now be processed near-instantaneously. The World Bank (2024) estimates that the cost of sending remittances globally has been falling steadily as digital alternatives displace traditional wire transfer services. By 2030, approximately USD \$25 trillion in consumer shopping value is expected to flow through mobile devices (Juniper Research, 2024), making digital platforms the dominant channel through which money moves globally.

The scale of disruption is illustrated by the growth of firms such as Revolut, a global financial application offering banking, cryptocurrency trading, insurance, and international transfers to over 65 million customers across 100 countries, with 2024 revenue of £3.1 billion — a 72% increase on its 2023 revenue (Revolut, 2025). Similarly, Checkout.com processes over £220 billion in annual e-commerce payment volumes for clients including Netflix, eBay,

and ASOS (Checkout.com, 2024). These figures rival those of established mid-sized national banking systems, underscoring the magnitude of disruption underway.

Valdez and Molyneux (2016) describe the deposit-taking institution as the cornerstone of the financial system: accepting short-term deposits, transforming them into longer-term loans, and thereby performing the maturity transformation that lubricates economic activity. This model is now being tested from multiple directions simultaneously. On the liabilities side, challenger banks and e-money institutions compete for deposits through superior digital interfaces and more attractive savings rates. On the assets side, fintech lenders and peer-to-peer platforms are eroding traditional loan markets by delivering faster credit decisions and lower prices, underpinned by algorithms rather than branch-based relationship managers.

Digital currencies — whether privately issued cryptocurrencies or central bank digital currencies (CBDCs) — deepen this disruption. Blockchain technology enables transactions to be recorded on a decentralised ledger without a central clearing authority, thereby reducing settlement time and counterparty risk (Saunders, Cornett and Erhemjamts, 2023). More significantly, digital currencies introduce the possibility of disintermediation: the bypassing of traditional bank intermediaries altogether, enabling savers and borrowers, or payers and payees, to interact directly without institutional mediation (Mishkin and Eakins, 2024; BIS, 2023).

Historically, asymmetric information confined low-income and unbanked populations to informal credit markets or excluded them from formal finance entirely (Mishkin and Eakins, 2024). Fintech firms are beginning to address this structural failure. Abound, a UK-based AI-powered lender, uses open banking data to assess affordability and extend fair credit to borrowers underserved by conventional scoring methods, with cumulative loan issuance reaching £900 million and revenue growing 151% in FY2025 (Abound, 2025). This exemplifies how data-driven intermediation can widen access to financial services without the legacy cost structures of traditional banks.

The evolution of the sector can be traced through three distinct phases. Prior to the 2008 Global Financial Crisis (GFC), financial intermediation was dominated by large universal banks operating through extensive branch networks, with payment systems characterised by batch-processing infrastructure such as the Bankers' Automated Clearing Service (Bacs). The post-crisis period (2009–2016) saw public trust in traditional banks severely damaged and regulatory capital requirements tightened under Basel III, creating space for a generation of

fintech start-ups that often benefitted from lighter regulatory treatment. From 2017 onwards, the introduction of open banking regulation — PSD2 in the European Union, subsequently adopted in the UK — compelled banks to share customer data with authorised third parties via application programming interfaces (APIs), dramatically lowering barriers to entry for fintech competitors and accelerating market disruption (Valdez and Molyneux, 2016).

Legacy infrastructure has also begun to adapt. The UK's Faster Payments System (FPS) processed over 5.09 billion transactions in 2024 with a total value of £4.2 trillion, while the Bacs system handled 6.86 billion transactions in the same period (Pay.UK, 2025). The UK government has identified fintech as a national strategic priority: the Chancellor of the Exchequer, Rachel Reeves, made clear in her 2024 Mansion House speech the fundamental role of financial services — including the fintech sector — in the UK's broader growth ambitions (HM Treasury, 2024). The theoretical frameworks of financial intermediation, maturity transformation, and payments architecture — developed over decades in academic and policy discourse, as articulated by Mishkin and Eakins (2024), Valdez and Molyneux (2016), and Saunders et al. (2023) — are therefore being stress-tested in real time by a wave of technological innovation that shows few signs of abating.

II. Critical Review of the Academic and Policy Literature

The four sources reviewed below have been selected to provide complementary perspectives on fintech's impact on financial intermediation and payment systems. Together they span quantitative empirical analysis of US credit markets, geographic financial inclusion research, and two major central bank and international policy documents on CBDCs — offering a range of methodological approaches and institutional vantage points from which to evaluate the transformation underway.

Source 1: Buchak, G., Matvos, G., Piskorski, T. and Seru, A. (2018) 'Fintech, Regulatory Arbitrage, and the Rise of Shadow Banks', Journal of Financial Economics, 130(3), pp.453–483.

Buchak et al. (2018) present one of the most influential empirical studies of fintech's impact on credit markets. Using a dataset of over 5.4 million US mortgage originations between 2010 and 2016, the authors document that fintech lenders' share of mortgage origination grew from approximately 2% to 8% over this period. Two principal mechanisms drive this growth: regulatory arbitrage — fintech lenders operating outside the scope of Basel III capital requirements enjoy a structural cost advantage — and technological efficiency gained through automated underwriting, algorithmic pricing, and fully digital origination. The use of granular

loan-level data and structural modelling allows the authors to decompose these two effects separately, which is a significant methodological strength. However, the analysis is confined to the US residential mortgage market and predates open banking regulation, meaning findings may understate the technological efficiency gains now achievable under contemporary data-sharing regimes.

In relation to the module, this source directly tests the financial intermediation theory described by Mishkin and Eakins (2024), who argue that banks' comparative advantage in information gathering justifies their dominance in credit markets. Buchak et al. (2018) demonstrate that this advantage is contestable when the regulatory framework is asymmetric, confirming Valdez and Molyneux's (2016) observation that competitive pressures from non-bank entrants can structurally reshape banking markets. The finding that regulatory arbitrage is a primary driver also raises the systemic risk concerns highlighted by Saunders et al. (2023): credit risk may be migrating outside the regulated perimeter in ways that are difficult for supervisors to monitor, a challenge that central bank mandates as discussed in the module are not yet fully equipped to address.

Source 2: Jagtiani, J. and Lemieux, C. (2018) 'Do Fintech Lenders Penetrate Areas That Are Underserved by Traditional Banks?', Journal of Economics and Business, 100, pp.43–54.

Jagtiani and Lemieux (2018) offer a complementary perspective by focusing on the financial inclusion dimension of fintech lending. Using loan-level data from LendingClub — then the world's largest peer-to-peer lending platform — the authors find that fintech lenders serve geographic markets and demographic segments that traditional banks have under-served or exited following the GFC. Crucially, fintech lenders use alternative data sources beyond standard credit bureau scores — including rental payment histories, utility bills, and online behavioural data — to extend credit to borrowers who would have been declined by conventional underwriting. A key strength is the paper's demonstration that this expansion reflects genuine market gap-filling rather than cherry-picking of low-risk borrowers. Limitations include the single-platform dataset and LendingClub's subsequent transition to a bank charter in 2021, which reduces the direct applicability of findings to present-day platform lending markets.

The paper maps directly onto Mishkin and Eakins' (2024) analysis of how asymmetric information has historically excluded low-income households from affordable credit. Fintech's use of alternative data is shown empirically to mitigate this informational barrier, extending

intermediation to previously excluded groups and supporting Valdez and Molyneux's (2016) broader view that the evolution of financial markets has consistently expanded access, albeit unevenly. Together, Buchak et al. (2018) and Jagtiani and Lemieux (2018) present a nuanced picture: fintech simultaneously disrupts established institutions and fills the gaps those institutions leave behind, challenging simple narratives of either pure disruption or pure democratisation.

Source 3: BIS (2021) 'CBDCs: An Opportunity for the Monetary System', BIS Annual Economic Report 2021, Chapter III. Basel: Bank for International Settlements.

The BIS (2021) Annual Economic Report examines the design and implications of CBDCs using survey evidence from 65 central banks collected in 2020. It finds that 86% of central banks were actively researching CBDCs, 60% were in experimental work, and 14% were running pilot projects. The BIS argues that well-designed CBDCs can address key deficiencies in the monetary system — high cross-border payment costs, financial exclusion, and instability risks from private stablecoins — while preserving the two-tier structure in which central banks and commercial banks occupy complementary roles. The report's combination of global survey evidence and monetary system modelling gives it broad credibility. However, having been published before widespread CBDC deployments had generated observable data, its projections remain empirically unvalidated.

The disintermediation risk that the BIS (2021) report explicitly flags — that depositors might shift funds from commercial banks to CBDC accounts at scale — resonates directly with Mishkin and Eakins' (2024) discussion of commercial bank deposits as the foundation of credit creation and monetary transmission. If CBDCs erode the deposit base, the central bank's ability to influence the economy through conventional policy channels could be impaired, a concern echoed in Saunders et al. (2023). The report therefore extends rather than challenges the module's theoretical framework, affirming the continuing relevance of central bank mandates while identifying the institutional challenges that digital innovation creates for policymakers.

Source 4: Bank of England and HM Treasury (2023) 'The Digital Pound: A New Form of Money for Households and Businesses', Consultation Paper. London: Bank of England.

The Bank of England and HM Treasury (2023) consultation paper provides the most direct UK-specific policy treatment of retail CBDC. It proposes a digital pound accessible via digital wallets, operating alongside — rather than replacing — existing money. The paper examines programmability, interoperability with private payment systems, privacy protections, and

holding limits, proposing an initial individual cap of £10,000–£20,000 specifically to mitigate the risk of large-scale commercial bank disintermediation. As a policy consultation, its methodology relies on scenario analysis and stakeholder engagement rather than quantitative modelling, limiting empirical rigour; the Bank of England has itself acknowledged the need for further research on disintermediation scenarios before a final CBDC decision is reached.

The consultation paper directly operationalises the central bank mandates — price stability, financial stability, and payment system oversight — discussed throughout the module, illustrating how these mandates must be reinterpreted in a digital environment. The proposed holding limits reflect the tension identified by Valdez and Molyneux (2016) between central bank control of the monetary architecture and the commercial banking sector's need for stable deposit funding. The paper raises regulatory design questions that mirror Saunders et al.'s (2023) analysis of prudential supervision in an era of rapid innovation. Taken with the BIS (2021) report, it suggests that, while the case for exploring CBDCs is broadly accepted among central banks, the practical design trade-offs remain deeply contested and the institutional stakes are high.

Collectively, these four sources present a complementary picture. Buchak et al. (2018) and Jagtiani and Lemieux (2018) establish that fintech disruption in credit markets is quantifiable and driven by both regulatory asymmetry and data-enabled efficiency. The BIS (2021) and Bank of England (2023) shift focus to the institutional response, showing how central banks are working to reclaim the monetary architecture that private digital innovation threatens to reshape. The tension between private innovation and public oversight will define the trajectory of financial intermediation for the coming decade.

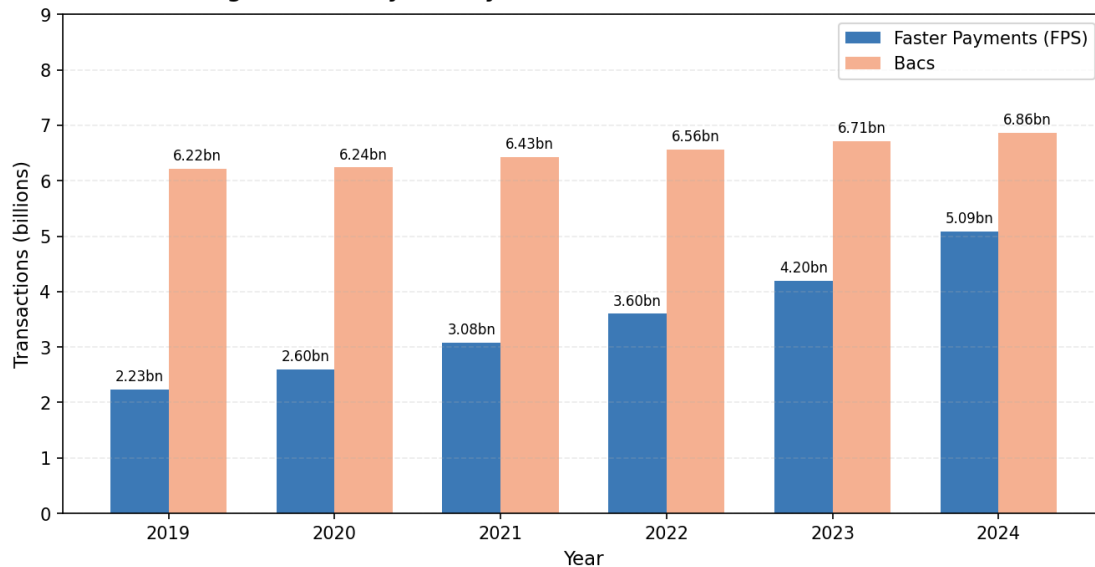
III. Real-Time Data Exercise and Conclusions

This section presents original analysis of two publicly available datasets to evaluate the empirical reality of fintech's impact on payment systems and financial inclusion. The first dataset is drawn from Pay.UK's annual payment clearing statistics; the second from the World Bank Global Findex Database 2025, which reports data collected in 2024 across more than 100 economies. Figures 1 and 2 present the findings visually, supported by the data tables below.

Table 1: UK Retail Payment System Activity, 2019–2024

Year	FPS Transactions (bn)	FPS Value (£tn)	Bacs Transactions (bn)	Bacs Value (£tn)
2019	2.23	1.4	6.22	4.9
2020	2.60	1.9	6.24	5.0
2021	3.08	2.4	6.43	4.7
2022	3.60	2.8	6.56	4.8
2023	4.20	3.5	6.71	4.9
2024	5.09	4.2	6.86	5.1

Source: Pay.UK (2024, 2025). FPS = Faster Payments System; Bacs = Bankers' Automated Clearing Service. Values are approximate and subject to revision.

Figure 1: UK Payment System Transaction Volumes (2019–2024)

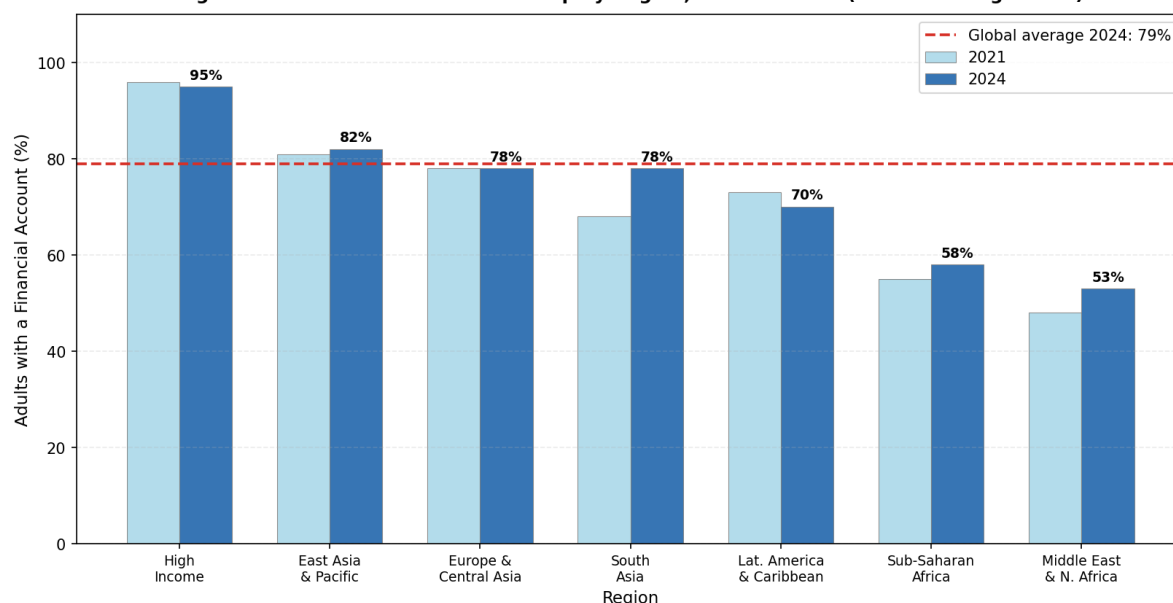
Source: Pay.UK (2024, 2025). FPS transaction volume growth reflects the accelerating shift to real-time digital payments; Bacs shows near-flat growth, confined to recurring payments.

The data in Table 1 and Figure 1 reveal a structural shift in UK retail payment infrastructure. Faster Payments transactions nearly doubled between 2019 and 2024 — from 2.23 billion to 5.09 billion — reflecting the rapid migration of payments from slower batch-processing systems to real-time digital rails, driven by mobile banking adoption, open banking, and the growth of e-commerce. The value of FPS transactions rose from £1.4 trillion to £4.2 trillion over the same period, confirming that real-time payments are not confined to low-value consumer flows but increasingly underpin significant commercial activity. By contrast, Bacs volumes grew modestly (from 6.22 billion to 6.86 billion), reflecting its continued role in recurring salary and direct debit payments. These divergent trajectories are consistent with the disruption narrative advanced by Buchak et al. (2018): established infrastructure, like established credit intermediaries, is being challenged by faster, more flexible digital alternatives that better match the expectations of a digitally native population.

Table 2: Global Financial Inclusion and Mobile Money Adoption, 2011–2024

Indicator	2011	2014	2017	2021	2024
Global account ownership (% of adults)	51%	62%	69%	74%	79%
Mobile money accounts – Global (%)	<1%	2%	5%	11%	16%
Mobile money – Sub-Saharan Africa (%)	<1%	12%	21%	33%	40%
Mobile money – Latin America & Caribbean (%)	<1%	4%	8%	24%	37%
Mobile money – South Asia (%)	<1%	3%	6%	12%	22%
Financially excluded adults (approx. billions)	~2.5	~2.0	~1.7	~1.4	~1.3

Source: World Bank Global Findex Database 2025 (2024 survey wave). Mobile money = personally using a mobile phone to make or receive payments. Findex 2025 prioritised low- and middle-income economies; comparable global digital payment series unavailable for 2024. Unbanked figures are estimates based on country-level population-weighted data.

Figure 2: Financial Account Ownership by Region, 2021 vs. 2024 (% of adults aged 15+)

Source: World Bank Global Findex Database 2025 (2024 survey wave). Population-weighted regional averages. The dashed red line shows the global average of 79% in 2024.

The World Bank Findex 2025 data in Table 2 and Figure 2 document continued, if uneven, progress in global financial inclusion. The share of adults holding a financial account rose from 51% in 2011 to 79% in 2024 — a 28 percentage point increase over thirteen years. Progress between 2021 and 2024 has been more modest (a 5 percentage point gain), suggesting that the most straightforward gains from digital account expansion have already been captured, and that the remaining unbanked population faces deeper structural barriers. Mobile money has nonetheless continued to expand rapidly: globally, mobile money account ownership rose from 11% in 2021 to 16% in 2024. Sub-Saharan Africa remains the standout region, with 40% of adults holding a mobile money account in 2024 — up from 33% in 2021 — driven overwhelmingly by fintech platforms operating without traditional bank branch infrastructure.

Latin America and the Caribbean has also seen dramatic growth (24% in 2021 to 37% in 2024), reflecting the rapid scaling of platforms such as Nubank and Mercado Pago across the region.

Despite this progress, Figure 2 reveals that profound regional disparities persist in 2024. Account ownership in the Middle East and North Africa stands at 53%, and Sub-Saharan Africa at 58%, compared to 95% in high-income economies — a gap of over 37 percentage points that has narrowed only marginally since 2021. Approximately 1.3 billion adults remain financially excluded globally, a figure that has declined only modestly from an estimated 1.4 billion in 2021. This underscores that technology alone is insufficient to overcome the barriers of low income, limited digital connectivity, and inadequate identification infrastructure. Importantly, the Findex 2025 methodology adjusted its scope, asking financial use questions only in low- and middle-income economies, which precludes direct global comparison of digital payment adoption with earlier waves — a limitation worth noting when interpreting these data (World Bank, 2025). Together, these findings validate Jagtiani and Lemieux's (2018) inclusion thesis at a global scale while also confirming that the frontier of financial inclusion now lies in addressing structural barriers beyond the reach of digital product design alone.

Conclusions and Policy Implications

The evidence from both the academic literature and real-time data confirms that fintech and digital currencies are fundamentally reshaping deposit-taking institutions and payment systems, simultaneously expanding financial access and introducing new structural risks. Traditional banks retain significant advantages in regulatory trust, balance-sheet capacity, and institutional relationships. However, their comparative advantage in information intermediation — the very justification for their existence articulated by Mishkin and Eakins (2024) — is being progressively eroded by algorithmic alternatives that process data faster, at lower cost, and with greater flexibility. The BIS (2021) and Bank of England (2023) frameworks suggest that CBDCs may ultimately redefine the architecture of the monetary system itself, though the design choices and the distribution of risks remain deeply contested.

Two practical implications emerge. First, for regulators, the adoption of a "same activity, same regulation" principle — whereby fintech lenders and digital asset providers face equivalent requirements to traditional banks for equivalent functions — is essential to prevent the regulatory arbitrage documented by Buchak et al. (2018) and to ensure that systemic risk does not accumulate unobserved outside the regulated perimeter (Saunders, Cornett and Erhemjamts, 2023). The Financial Stability Board's ongoing work on regulatory frameworks

for fintech and crypto-assets represents meaningful progress, but cross-jurisdictional coordination remains incomplete, creating persistent opportunities for arbitrage.

Second, for traditional financial institutions, investment in digital infrastructure and open-API banking has become a competitive necessity rather than a strategic option. The UK's Faster Payments infrastructure provides a high-quality foundation, but institutions must build differentiated value on top of it — through personalisation, embedded finance propositions, and data-driven credit products — to retain relevance in a system where barriers to entry in payments have fundamentally collapsed. Regulatory risk — the prospect of tightening the asymmetric treatment currently enjoyed by non-bank lenders — represents the most significant medium-term uncertainty for investors and warrants close monitoring.

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